BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 005 · Floating point → denary

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Represent 6.5 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Q2. Convert $6D_{16}$ to eight-bit binary and denary.

Q3. Convert 11110110 from 8-bit two’s-complement binary to denary. Show signed place values.

Q4. Apply AND 00001111 to 10110110 . Give the result in binary and hexadecimal.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — Mantissa

Recall the term — Exponent

Recall the term — Normalised

Complete the statement — Precision

Complete the statement — Range

Recall the representation — Two’s complement

New learning

Split the fields — Separate the bit pattern into the mantissa and the exponent at the stated field boundary.

Evaluate the mantissa — Add the place values for the bits that are 1. — CODE:
Mantissa places:
−1 | ½ ¼ ⅛ 1⁄16 1⁄32 1⁄64 1⁄128

Convert the exponent to denary — Read the exponent as a signed two’s-complement integer . — CODE:
4-bit exponent places:
−8 4 2 1

Apply the scale — CODE:
Value = mantissa × 2^{E} — A positive exponent multiplies by a power of two. A negative exponent divides by a power of two.

Check the representation — For a non-zero normalised mantissa: — 01 starts a positive normalised mantissa.

Worked example

Convert this floating-point representation to denary

1. Evaluate the mantissa — CODE:
01011000
= ½ + ⅛ + 1⁄16
= 0.6875 — Read the mantissa as a signed binary fraction.

2. Convert the exponent to denary — CODE:
0011 = 3 — The exponent is a signed two’s-complement integer.

3. Apply the scale — CODE:
0.6875 × 2^{3}

= 0.6875 × 8
= 5.5

We Do · Practise with guidance

Convert this floating-point representation to denary

Convert this floating-point representation to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Build the answer

Convert to denary, then write in normalised floating-point form

Convert the value to denary, then write the same value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Shift the mantissa left twice and reduce the exponent by 2.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Convert 0100 1110 to denary
The mantissa is four bits and the exponent is four bits. Both use two’s complement.
Show your working.
Paraphrased OCR June 2023 Q3(d); original values and marks retained. Source .

Question 2

Convert 1010 1110 to denary
Use a four-bit mantissa and four-bit exponent. Show each field value, the scaling operation and the answer.
Paraphrased OCR June 2022 Q3(b); original values and marks retained. Source .

Question 3

Which representations are normalised?
Each representation uses a six-bit two’s-complement mantissa followed by a three-bit two’s-complement exponent. State which representations are normalised by checking the mantissa’s leading bits.
CODE:
a) 010101 100
b) 110101 111
c) 011010 010
d) 101010 110
Paraphrased OCR June 2022 Q3(c); original marks retained. Source .

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Question 4

Convert both floating-point representations to denary
(a) 01110000 0010 (b) 10010000 1111
Each representation uses an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Convert both to denary and show the mantissa value, exponent value and final denary result.

Question 5

Precision and range
What improves if more bits are allocated to (a) the mantissa and (b) the exponent?
Then explain the trade-off if the total number of bits is fixed.
Think about what mantissa bits control and what exponent bits control.
The first two prompts paraphrase OCR June 2023 Q3(e). The trade-off prompt is an original classroom extension. Source .

Exit Ticket · Answer before checking

A negative exponent means the floating-point value must be negative.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 01101000 0011 6.5 = 110.1_2 = $0.1101_2 \times 2^3$. Mantissa 01101000 ; exponent 0011 .

Q2. 01101101_2 and 109 . 6 → 0110 , D → 1101 . In denary: (6 × 16) + 13 = 109 .

Q3. −10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

Q4. 00000110_2 = 06_{16} AND with 00001111 clears the upper nibble and preserves the lower nibble 0110 .

You Do pages · Independent questions

Q1. CODE:
Mantissa 0100 = 0.5
Exponent 1110 = -2

0.5 × 2^-2 = 0.125

Answer = 0.125

Q2. CODE:
Mantissa 1010 = -0.75
Exponent 1110 = -2

-0.75 × 2^-2 = -0.1875

Answer = -0.1875

Q3. CODE:
a) 010101 100 → normalised
b) 110101 111 → not normalised
c) 011010 010 → normalised
d) 101010 110 → normalised

Rule:
positive normalised mantissa begins 01

negative normalised mantissa begins 10

Q4. CODE:
(a) 01110000 0010

mantissa = 0.1110000_2 = 0.875
exponent = 2

0.875 × 2^2 = 3.5

(b) 10010000 1111

mantissa = -1 + 1/8 = -0.875
exponent = -1

-0.875 × 2^-1 = -0.4375

Q5. CODE:
More mantissa bits:
→ greater precision / more significant binary digits.

More exponent bits:
→ greater range / larger and smaller magnitudes can be represented.

If the total number of bits is fixed:
allocating more bits to the mantissa improves precision but leaves fewer bits for the exponent and reduces range;
allocating more bits to the exponent increases range but leaves fewer bits for the mantissa and reduces precision.

Exit Ticket · Model answer

False. An exponent changes the magnitude; the sign of the mantissa determines whether the value is positive or negative.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: